Yiyang (Yvonne) Sun

647-986-3588
yvonnesunyy@gmail.com GitHub Website LinkedIn

Summary

Technical solutions engineer with experience leading credit risk and regulatory platforms within enterprise cloud environments. Proven ability to translate quantitative modelling workflows into scalable architectures by collaborating closely with business stakeholders, risk teams, and engineering partners. Strong background in system design, data platforms, and AI-informed solution development, focused on bridging models, infrastructure, and real-world decision workflows.

Technical Expertise

Cloud Platforms	Azure, GCP, AWS, Kubernetes, Docker, CI/CD, Terraform
Data & AI Systems	Python, SQL, Data Pipelines, Risk Modelling Ecosystems (SAS/R), Semantic & AI Workflows, Blockchain Concepts (Solidity)
Enterprise Applications	React, Angular, TypeScript, REST APIs, Workflow Systems, Azure AD Integration
Engineering Foundations	Java, C#, C++, System Design, Distributed Processing

Experience

Scotiabank, Toronto, ON

Manager, Retail Provisions Analytics — Credit Risk Solutions Jun 2025 – Present

- Supported monthly *Expected Credit Loss (ECL)* production for retail portfolios by bridging quantitative risk modelling, business reporting, and cloud-based data processing workflows.
- Collaborated with local business and risk stakeholders to present ECL outputs, explain model behaviour, and translate regulatory requirements into scalable technical solutions.
- Partnered closely with enterprise IT and platform engineering teams to redesign legacy **SAS**/**R** risk models into modular **Python** pipelines, enabling deployment within the bank's **GCP**-based processing framework.
- Architected large-scale credit risk computation pipelines processing **2.5M+** accounts monthly, improving execution efficiency by **40%** while enhancing governance transparency and operational resilience.

Software Engineer Jan 2025 – Jun 2025

- Partnered with business stakeholders to design end-to-end financial reporting and attestation solutions, translating governance requirements into scalable web platform architecture.
- Led frontend architecture design using **React (TypeScript)** and collaborated with database teams to define data access patterns and implement optimized **stored procedures**, enabling reliable cross-department workflows.
- Designed hierarchical workflow engines and tree-structured data visualizations, improving operational transparency and decision-making efficiency by **50%**.
- Integrated role-based identity management using **Azure AD (MSAL)** and coordinated with platform engineering teams to deploy solutions through **Azure**-based **CI/CD** pipelines.

iNAGO Corp, Toronto, ON

Software Engineer(Capstone Project) Jun 2021 – Apr 2022

- Designed and delivered a conversational route-planning platform translating user intent into optimized travel solutions through AI-assisted workflows.
- Architected backend services using **Python Django** and integrated optimization logic (**A* pathfinding**) into scalable RESTful APIs, enabling dynamic solution generation based on user constraints.
- Integrated **Google Dialogflow** to enable natural language interaction, achieving **85%** intent recognition accuracy while bridging conversational AI with structured application workflows.
- Designed system state management and decision orchestration logic to dynamically recompute routes as user preferences evolved, improving responsiveness and solution adaptability.

Bombardier Inc, Toronto, ON

Solution Engineer Intern Jun 2020 – May 2021

- Designed and delivered an internal aircraft workshop automation platform (*ePCS*) used daily by **150+** operators, translating operational workflows into scalable digital solutions.
- Collaborated with workshop stakeholders to gather requirements and define system architecture, designing a normalized **SQL Server** data model (**BCNF**) to ensure data integrity and long-term maintainability.
- Developed end-to-end application architecture using **C# ASP.NET** services and **Angular** frontend, implementing role-based access control to support multiple operational user groups.
- Optimized system performance and workflow efficiency through architectural refactoring, reducing latency by **15%** and improving operational throughput by **90%**.

Education

University of Toronto Sep 2022 - Jun 2024
M.A.Sc. in Industrial Engineering
Focus: Semantic Technologies, Knowledge Graphs, Applied Machine Learning, Visual Question Answering
Thesis: Semantic modeling of temporal/spatial structures (grid lattices) with applications in navigation and decision-support.

University of Toronto Sep 2017 - May 2022
B.A.Sc. in Industrial Engineering, Minor in Artificial Intelligence, Minor in Business